VALUATIVE EXISTENCE FOR THE PICARD STACK OF A SMOOTH PROPER FAMILY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1538-1544 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 1546-1568, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 16 of queue_moduli.json (pb=1083, stmt 1538-1544, proof 1546-1568).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DMM: sources/stacks-moduli/pinned-original-source.tex, source lines 1568–1568. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DMC: sources/stacks-moduli/pinned-original-source.tex, source lines 1568–1568. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04ZL: sources/stacks-moduli/pinned-original-source.tex, source lines 1568–1568. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06M1: sources/stacks-moduli/pinned-original-source.tex, source lines 1568–1568. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CLK: sources/stacks-moduli/pinned-original-source.tex, source lines 1568–1568. Cited by the retained statement or proof but

Date: stacks-project:moduli.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Assume $X \to B$ is smooth in addition to being proper. Then $\mathcal{P}ic_{X/B} \to B$ satisfies the existence part of the valuative criterion (Morphisms of Stacks, Definition 0CLK).*

Proof. Taking base change, this immediately reduces to the following problem: given a valuation ring R with fraction field K and an algebraic space X proper and smooth over R and an invertible $\mathcal{O}_{X_K}$-module $\mathcal{L}_K$, show there exists an invertible $\mathcal{O}_X$-module $\mathcal{L}$ whose generic fibre is $\mathcal{L}_K$. Observe that X_K is Noetherian, separated, and regular (use Morphisms of Spaces, Lemma 04ZL and Spaces over Fields, Lemma 06M1). Thus we can write $\mathcal{L}_K$ as the difference in the Picard group of $\mathcal{O}_{X_K}(D_K)$ and $\mathcal{O}_{X_K}(D'_K)$ for two effective Cartier divisors D_K, D'_K in X_K, see Divisors on Spaces, Lemma 0DMM. Finally, we know that D_K and D'_K are restrictions of effective Cartier divisors $D, D' \subset X$, see Divisors on Spaces, Lemma 0DMC. $\square$